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ACME PETROCHEMICAL DEMO COMPLEX - UNIT 2300

PROCESS DATA SHEET: COLUMN / VESSEL

Document Code: DS-V2301 | Revision: Z1 | Status: Issued for Construction

Equipment Tag: V-2301

Equipment Name: Preflash Column

Equipment Class: Distillation Column / Vacuum Flash Vessel

Process Unit: Unit 2300 (Concentration & Decomposition Section)

Service: Vacuum Concentration of Technical Cumene Hydroperoxide (CHP) Feed from 28 wt% to 82 wt%

#### 1. MECHANICAL & DESIGN DATA

- Internal Design Pressure: 3.5 kg/cm2g (Full Vacuum FV to +3.5 kg/cm2g)
- Operating Pressure (Top / Bottom): 35 mmHgA / 48 mmHgA (0.046 kg/cm2a)
- Internal Design Temperature: 165 degC
- Normal Operating Temperature: 92 degC (Bottoms) / 68 degC (Overhead Vapor)
- Inside Diameter (ID): 2100 mm
- Tangent-to-Tangent Length (T/T): 14500 mm
- Wall Thickness: 18 mm (including 3.0 mm Corrosion Allowance)
- Shell & Head Material: SA-516 Gr.70 Cladded with 3.0 mm SA-240 Type 316L
- Internals Material: SA-240 Type 316L Structured Packing (2 Beds, Mellapak 250Y)

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DS-V2301 PAGE 2 OF 9 - PROCESS OPERATING CONDITIONS & FLUID PROPERTIES

Equipment Tag: V-2301 (Preflash Column)

Feed Stream S210 (Dilute CHP from Oxidation):

- Mass Flow Rate: 42,500 kg/hr
- Operating Temperature: 78 degC
- Operating Pressure: 2.4 kg/cm2g
- Composition: 28.0 wt% Cumene Hydroperoxide (CHP), 68.5 wt% Cumene, 2.5 wt% DMBA, 1.0 wt% Acetophenone

Bottoms Stream S229 (Concentrated CHP to Decomposer Reactor D-2304 via Pump P-2301A/B):

- Mass Flow Rate: 14,500 kg/hr
- Operating Temperature: 92 degC (Max Permitted Skin Temp: 105 degC due to CHP SADT)
- Composition: 82.0 wt% Cumene Hydroperoxide (CHP), 13.0 wt% Cumene, 3.8 wt% DMBA, 1.2 wt% Acetophenone

Overhead Vapor Stream S220 (Recovered Cumene Vapor to Vacuum System X-2301):

- Mass Flow Rate: 28,000 kg/hr at 68 degC and 35 mmHgA

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Equipment Tag: V-2301

| Nozzle | Size  | Rating | Facing | Service                         | Connected Line / Stream                   |
|--------|-------|--------|--------|---------------------------------|-------------------------------------------|
| N1     | 6 in  | 150#   | RF     | Dilute CHP Feed Inlet           | 6"-P-2301-316L (Stream S210)              |
| N2     | 4 in  | 150#   | RF     | Concentrated CHP Bottoms Outlet | 4"-P-2302-316L (Stream S229 to P-2301A/B) |
| N3     | 12 in | 150#   | RF     | Overhead Vapor Outlet           | 12"-P-2303-316L (Stream S220 to X-2301)   |
| N4     | 3 in  | 150#   | RF     | Reflux Return Inlet             | 3"-P-2304-316L                            |
| N5     | 4 in  | 150#   | RF     | Relief Valve Connection         | PSV-2301 (Set 3.5 kg/cm2g to D-2320)      |
| N6     | 2 in  | 150#   | RF     | Emergency Cumene Flush Inlet    | 2"-P-2309-316L (Interlock I-2301)         |

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Equipment Tag: V-2301

- Rectifying Bed (Bed #1): Structured Packing 316L, Height 3200 mm, HETP 400 mm
- Stripping Bed (Bed #2): Structured Packing 316L, Height 4500 mm, HETP 420 mm
- Liquid Distributor: Low-Holdup Trough Distributor Type LD-316L (Residence time < 45 seconds)
- Design Note: Minimum liquid holdup in V-2301 sump is mandatory to prevent thermal decomposition of CHP.

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Equipment Tag: V-2301

- Pressure Transmitter PT-0401: Range 0 to 100 mmHgA, controls Vacuum Control Valve PV-0401
- Feed Flow Transmitter FT-0401: Coriolis Mass Flow Meter on Stream S210 (0 to 60,000 kg/hr)
- Sump High Temperature Trip TAHH-0402: Setpoint 105.0 degC triggers Interlock I-2301
- Pressure Safety Valve PSV-2301: Set Pressure 3.5 kg/cm2g, Orifice 3K4, discharges to D-2320

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DS-V2301 PAGE 6 OF 9 - MECHANICAL LOADS, WIND & SEISMIC DESIGN

Equipment Tag: V-2301

- Wind Design Code: ASCE 7-16 (Basic Wind Speed 40 m/s)
- Seismic Zone: Zone 2A (Importance Factor 1.25)
- Skirt Support: SA-516 Gr.70, Height 4200 mm, Fireproofed with 50 mm cementitious coating

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Equipment Tag: V-2301

- ASME Boiler and Pressure Vessel Code Section VIII Division 1
- Radiography: 100% RT on all butt welds (Full Vacuum & Lethal/Peroxide Service)
- Post-Weld Heat Treatment (PWHT): Required on Carbon Steel backing prior to 316L passivation

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DS-V2301 PAGE 8 OF 9 - INSULATION, PAINTING & SURFACE PREPARATION

Equipment Tag: V-2301

- Insulation: 65 mm Cellular Glass (Non-absorbent to prevent auto-ignition of organic peroxides)
- Warning: Mineral wool or fibrous silicate insulation is strictly prohibited in CHP service.

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DS-V2301 PAGE 9 OF 9 - REVISION HISTORY & QUALITY SIGN-OFF

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Prepared by: Acme Process Engineering Group

Approved for Unit 2300 Synthetic Benchmark Corpus.